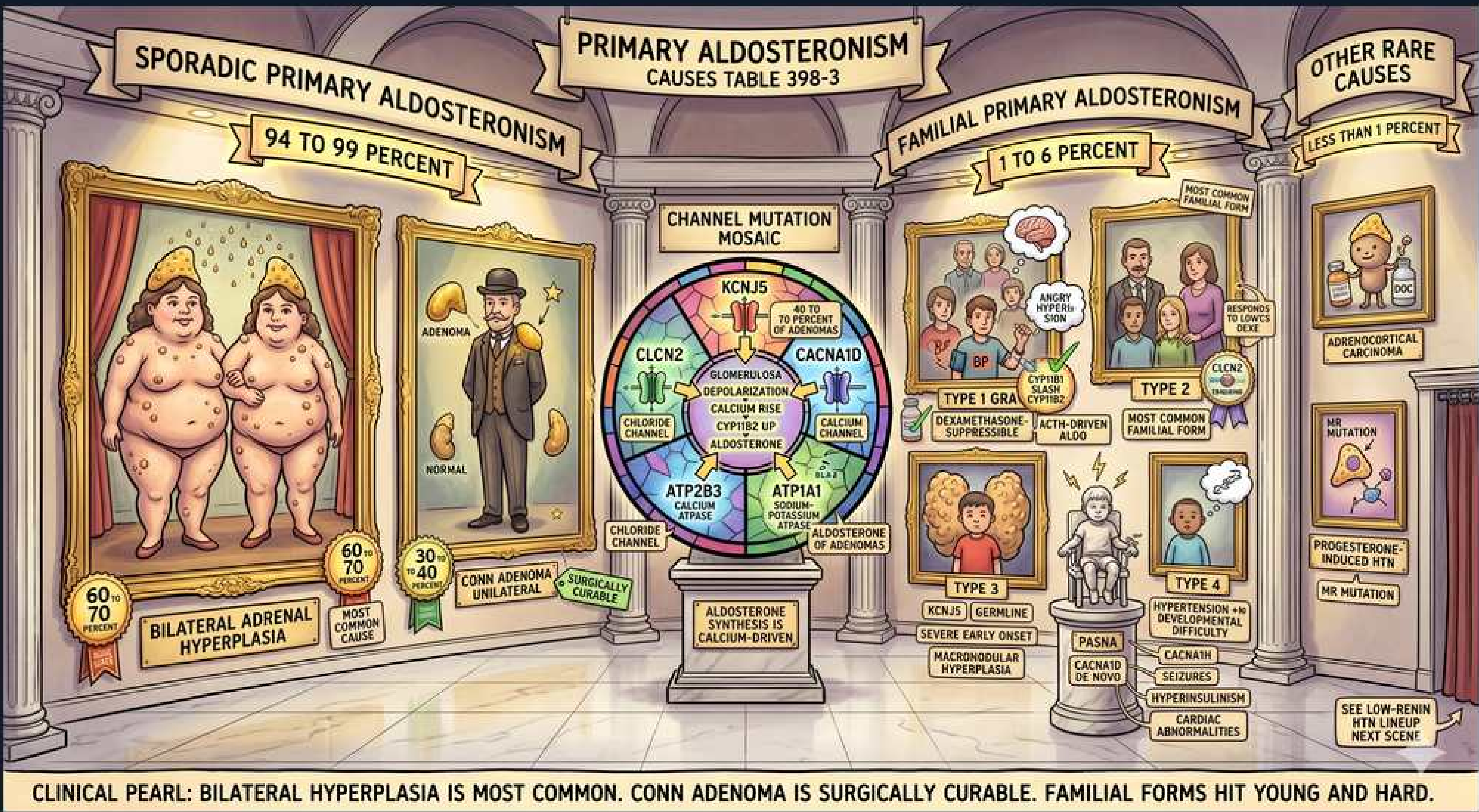
Primary Aldosteronism — The Aldosterone Causes Gallery



1. Sporadic = 94-99%

WHAT YOU SEE

Left archway SPORADIC PRIMARY ALDOSTERONISM 94 to 99 percent

WHAT IT ENCODES

The overwhelming majority of primary aldosteronism (94–99%) is SPORADIC, not inherited. Within sporadic disease the two big buckets are bilateral idiopathic hyperplasia and unilateral aldosterone-producing adenoma (Conn). The clinical hallmark of all PA: autonomous aldosterone → HTN, suppressed renin, ± hypokalemia/metabolic alkalosis.

BOARD TRAP

Familial forms are rare — don't over-attribute PA to genetics. But DO suspect a familial form in young/early-onset patients with a strong family history.

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2. Bilateral hyperplasia (most common)

WHAT YOU SEE

Framed portrait of two identical figures, BILATERAL ADRENAL HYPERPLASIA, 60-70%

WHAT IT ENCODES

Bilateral adrenal (idiopathic) hyperplasia is the SINGLE MOST COMMON cause of PA (~60–70%). Because BOTH glands over-secrete, surgery is not curative — management is lifelong medical therapy with a mineralocorticoid-receptor antagonist (spironolactone first-line, eplerenone if side effects).

BOARD TRAP

Sending bilateral hyperplasia to surgery is the classic error — it is managed MEDICALLY with an MRA. AVS (not CT) is what proves the disease is bilateral and spares the patient an unnecessary adrenalectomy.

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3. Conn adenoma (unilateral)

WHAT YOU SEE

Single dapper figure, CONN ADENOMA UNILATERAL, 30-40%

WHAT IT ENCODES

Aldosterone-producing adenoma (Conn syndrome) is unilateral (~30–40% of PA) and is SURGICALLY CURABLE by laparoscopic adrenalectomy after AVS confirms lateralization. Tends to present with more florid biochemistry (lower K+, higher aldosterone) than bilateral hyperplasia.

BOARD TRAP

A unilateral 'adenoma' on CT may be a non-functioning incidentaloma — lateralization must be confirmed by AVS before resection, especially in patients ≥35.

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4. Channel mutation mosaic

WHAT YOU SEE

Color wheel CHANNEL MUTATION MOSAIC: KCNJ5, CACNA1D, ATP1A1, ATP2B3, CLCN2

WHAT IT ENCODES

Somatic mutations in ion channels/pumps drive autonomous aldosterone synthesis by depolarizing the glomerulosa cell and raising intracellular calcium (the trigger for CYP11B2/aldosterone synthase). KCNJ5 (a GIRK potassium channel) is the MOST COMMON somatic mutation in adenomas; others: CACNA1D, ATP1A1, ATP2B3, CLCN2.

BOARD TRAP

These are SOMATIC (tumor) mutations in sporadic adenomas — distinct from the GERMLINE versions (e.g. germline KCNJ5 = FH type 3; germline CACNA1D = PASNA/FH type 4) that cause familial disease.

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5. Familial = 1-6%

WHAT YOU SEE

Right archway FAMILIAL PRIMARY ALDOSTERONISM 1 to 6 percent

WHAT IT ENCODES

Familial hyperaldosteronism is uncommon (1–6%) but important: it presents YOUNG, often with severe/resistant HTN and a strong family history of early HTN or hemorrhagic stroke. Four recognized types (FH-I through FH-IV).

BOARD TRAP

Consider genetic testing/familial PA in any patient with PA onset before ~20 or a family history of early-onset HTN or stroke — easy to miss if you only think 'adenoma vs hyperplasia.'

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6. Type 1 GRA

WHAT YOU SEE

TYPE 1 GRA family, dexamethasone suppressible

WHAT IT ENCODES

FH type 1 = glucocorticoid-remediable aldosteronism (GRA): an unequal crossover creates a CYP11B1/CYP11B2 CHIMERIC gene placing aldosterone synthase under ACTH control. Aldosterone is therefore ACTH-driven and SUPPRESSIBLE by low-dose dexamethasone (which lowers ACTH). Associated with early hemorrhagic stroke from cerebral aneurysms.

BOARD TRAP

GRA is the one PA treated with low-dose glucocorticoid (dexamethasone), NOT primarily an MRA — though an MRA can be added. Suspect it in young patients with a family history of early stroke.

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7. Type 2

WHAT YOU SEE

TYPE 2 family portrait, most common familial form

WHAT IT ENCODES

FH type 2 is the MOST COMMON familial form. It is clinically indistinguishable from sporadic PA (adenoma or hyperplasia), is NOT dexamethasone-suppressible, and is linked to germline CLCN2 mutations. Managed like sporadic PA (MRA or adrenalectomy depending on lateralization).

BOARD TRAP

FH-2 looks exactly like sporadic PA — the only clue is the family history; don't expect a dexamethasone response (that's FH-1/GRA).

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8. Type 3 (KCNJ5 germline)

WHAT YOU SEE
TYPE 3 child, KCNJ5 germline, severe early onset

WHAT IT ENCODES
FH type 3 = GERMLINE KCNJ5 mutation causing severe, very-early-onset PA with massive bilateral adrenal hyperplasia, often refractory to medical therapy and requiring bilateral adrenalectomy.

BOARD TRAP
Same gene (KCNJ5) as the common SOMATIC adenoma mutation, but germline → far more severe, childhood-onset, bilateral disease. Don't conflate the somatic and germline phenotypes.

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9. Type 4 / PASNA

WHAT YOU SEE
PASNA figure, CACNA1D, seizures, neuro abnormalities

WHAT IT ENCODES
FH type 4 / PASNA syndrome = germline CACNA1D (L-type calcium channel) mutation causing Primary Aldosteronism with Seizures and Neurologic Abnormalities (developmental delay, neuromuscular features).

BOARD TRAP
The neuro features (seizures, developmental delay) are the giveaway that aldosteronism is part of a germline channelopathy (CACNA1D), not isolated adrenal disease.

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10. Adrenocortical carcinoma

WHAT YOU SEE
OTHER RARE CAUSES, ADRENOCORTICAL CARCINOMA, less than 1 percent

WHAT IT ENCODES
Rare causes (<1%): an aldosterone-secreting adrenocortical CARCINOMA. Suspect ACC when the adrenal mass is large (>4–6 cm), heterogeneous, high-attenuation on CT, and/or co-secretes other steroids (cortisol, androgens).

BOARD TRAP
A large, hormonally mixed adrenal mass causing aldosterone excess is ACC until proven otherwise — don't assume a benign Conn adenoma based on the biochemistry alone.

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11. Ectopic / progesterone-induced

WHAT YOU SEE
Progesterone induced with MR mutation note

WHAT IT ENCODES
Very rare: a gain-of-function mineralocorticoid-receptor (NR3C2) mutation (Geller mutation) makes the MR aberrantly activated by PROGESTERONE, causing severe HTN that paradoxically WORSENS in pregnancy when progesterone surges.

BOARD TRAP
In this MR-mutation syndrome spironolactone is an MR AGONIST (not antagonist) and worsens HTN — a classic pharmacology twist; HTN flaring in pregnancy is the clue.

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12. Clinical pearl

WHAT YOU SEE
Bottom banner

WHAT IT ENCODES
One-line summary: bilateral idiopathic hyperplasia is the MOST COMMON cause (treat medically with an MRA); Conn adenoma is unilateral and surgically curable (after AVS); familial forms (FH 1–4) are rare but hit young and hard — remember GRA is dexamethasone-suppressible.

BOARD TRAP
Two highest-yield points: (1) most common = bilateral hyperplasia → MRA, not surgery; (2) the one familial form treated with glucocorticoid is FH-1 / GRA.
